

Question ID: 97631565

| Assessment | Test | Domain                            | Skill                                                | Difficulty                                        |
|------------|------|-----------------------------------|------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Inference from sample statistics and margin of error | Easy <div><div></div><div></div><div></div></div> |

Question

Scott selected **20** employees at random from all **400** employees at a company. He found that **16** of the employees in this sample are enrolled in exactly three professional development courses this year. Based on Scott’s findings, which of the following is the best estimate of the number of employees at the company who are enrolled in exactly three professional development courses this year?

Answer

- A. **4**
- B. **320**
- C. **380**
- D. **384**

Correct Answer: B

Rationale

Choice B is correct. It’s given that from the sample of **20** employees at the company, **16** of the employees are enrolled in exactly three professional development courses this year. Since  $\left(\frac{16}{20}\right)$  is equal to **0.80**, or  $\frac{80}{100}$ , it follows that **80%** of the employees in the sample are enrolled in exactly three professional development courses this year. Therefore, the best estimate for the percentage of employees at the company who are enrolled in exactly three professional development courses this year is **80%**. It’s given that there are a total of **400** employees at the company. Therefore, the best estimate of the number of employees at the company who are enrolled in exactly three professional development courses this year is  $\left(\frac{80}{100}\right)(400)$ , or **320**.

Choice A is incorrect. This is the number of employees from the sample who aren’t enrolled in exactly three professional development courses this year.

Choice C is incorrect. This is the number of employees who weren’t selected for the sample.

Choice D is incorrect and may result from conceptual or calculation errors.